Saket Sandeep Deshmukh

AI/GenAI Engineer

Pune, India — +91 7666114965 — heyiamsaket@gmail.com — linkedin.com/in/saket-deshmukh — github.io/MyPortfolio

SUMMARY

AI Engineer with **2.5+ years of experience** building production-grade **LLM, RAG, and Agentic AI systems**. Skilled in Python, FastAPI, LangChain, LangGraph, PostgreSQL, and AWS. Reduced query resolution time by **52%**, hallucinated responses by **38%**, and achieved **99% schema compliance**.

SKILLS

Programming: Python, Bash
Generative AI: LLMs, RAG, Agentic AI, Prompt Engineering, Grounding, Guardrails
LLM Engineering: LLM Evaluation, Structured Output, Function Calling, Embeddings
AI Frameworks: LangChain, LangGraph, LiteLLM, MCP
AI Platforms: Amazon Bedrock, OpenAI API, Ollama, DeepSeek API, vLLM
Vector Search & Retrieval: pgvector, FAISS, BM25, Hybrid Search, Semantic Search
Cloud & DevOps: AWS, Docker, Git, Kubernetes. GitHub Actions, Terraform, Ansible
Databases: PostgreSQL, MySQL

EXPERIENCE

KWAD Corporation Pvt. Ltd. – Software Engineer **Apr 2025 – Present**

AI Platform – Clinical Record Processing

- Built a **Python/FastAPI AI platform** for extracting structured clinical data from **5K+ scanned, handwritten, and spreadsheet IVF records**, reducing manual data-processing effort.
- Designed a multimodal document-processing pipeline covering API, S3, and SFTP ingestion, document classification, OCR, vision-model transcription, Pandas-based parsing, and multilingual processing.
- Implemented **AWS Bedrock LLM orchestration** with document-based model routing, Pydantic structured outputs, and validation retries, achieving **99% schema compliance**.
- Built a deterministic grounding layer to verify extracted fields against original document source spans, automatically validating **75% of fields** and routing exceptions for human review. Implemented RAG using AWS Bedrock Knowledge Bases, S3, Titan V2 embeddings, Nova 2 Lite, and OpenSearch Serverless.

Fintech – AI Compliance & Financial Operations Copilot

- Designed and developed a production-grade **agentic RAG system** for operations and compliance teams to investigate B2B transaction data across **12 tenants** using natural-language queries.
- Built a **LangGraph orchestration layer** supporting multi-step workflows, intent and risk classification, conditional routing, tool calling, state management, retries, and human escalation.
- Developed a RAG architecture using **PostgreSQL, pgvector, embeddings, and semantic search**, integrating structured transaction data with unstructured compliance evidence. Connected OpenAI GPT-4o and LiteLLM with transaction SQL tools and deterministic compliance rules.
- Implemented **FastAPI authentication and RBAC** with tenant-aware authorization, enforcing authenticated tenant IDs across SQL queries and vector searches. Added evidence aggregation, confidence/risk evaluation, source citations, and human review for ambiguous or high-risk investigations.
- Owned LLM evaluation and observability through offline test sets, tracing, telemetry dashboards, and automated regression checks. Completed **9 optimization iterations**, reducing hallucinated responses by **38%** and regression detection time by **45%**, while maintaining investigation traceability in a regulated financial environment.

Briowin Private Limited – Software Engineer **Feb 2024 – Mar 2025**

E-Commerce – AI-Powered Product Information Assistant

- Developed an AI-powered product information assistant for a **300+ SKU catalog**, reducing product query resolution time by **52%**.
- Built a **LangGraph agent workflow** with conditional routing, input validation, hybrid retrieval, and fallback handling, achieving a **94% end-to-end request success rate**.
- Integrated **MCP** for standardized tool-based communication between the LLM application and product catalog services.

Implemented hybrid retrieval using OpenAI text-embedding-3-small, BM25, and Reciprocal Rank Fusion, improving top-k retrieval accuracy by **27%**.

- Implemented input/output guardrails and an automated **RAG evaluation pipeline using TruLens** to measure groundedness, context relevance, and answer relevance.

PROJECTS

AI - SQL Query Optimizer *Python, FastAPI, PostgreSQL, SQLGlot, DeepSeek API, Pydantic, pytest*

- Built a SQL optimization tool using **SQLGlot**, PostgreSQL metadata, and `EXPLAIN` plans to detect query bottlenecks and generate LLM-powered optimization suggestions.
- Implemented a SQL safety pipeline with **AST-based validation**, read-only access, query timeouts, result comparison, and rollback-based index testing.
- Developed **FastAPI REST APIs**, a web UI, and automated benchmarking, achieving **99.8% faster lookups** and **54% faster aggregations**, with **49 automated tests** and Docker Compose.

Self-Hosted LLM Inference Platform *AWS EKS, Kubernetes, Terraform, Docker, vLLM*

- Designed a self-hosted LLM inference platform using vLLM, Docker, Kubernetes, and AWS EKS to support controlled, scalable model serving.

EDUCATION

B.E. in Computer Engineering – Savitribai Phule Pune University **2019 – 2023**

CERTIFICATIONS

- **Evaluating RAG and Semantic Search Systems** – NVIDIA
- **Prompt Engineering for Everyone** – Cognitive Class
- **LangChain Mastery** – Udemy
- **Python for Data Science** – edX

AWARDS

- **Product and Platforms Pinnacle Award** – KWAD Corporation, 2026